

# Suraj Chougule

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## SUMMARY

**Software Engineer** with **3 shipped production projects** achieving **77x latency improvement**, **1,500+ RPS throughput**, and serving **162+ organic users**. Expert in **distributed systems**, **high-performance caching**, and **scalable full-stack development** with **Java, Python, C/C++, JavaScript (React, Node.js)**, and **AWS**. **1st Place Hackathon Winner**. Passionate about building systems at scale with measurable impact and production-grade observability.

## EDUCATION

**Aug 2023 – May 2027** B.E. Information Technology, **I<sup>2</sup>IT Pune** **CGPA: 8.84/10**  
**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Object-Oriented Design, Computer Networks, Software Engineering, Distributed Systems

## PROFESSIONAL EXPERIENCE

### NoteShare — Production-Scale Academic Platform [\[Live\]](#) [\[GitHub\]](#)

- **Performance Optimization:** Achieved **77x latency improvement** (315ms → 4ms P95) and **1,500+ RPS** throughput using L1 RAM + L2 Redis caching under **100 concurrent users** — reducing database load by **90%**
- **Scale & Capacity:** Built distributed MERN backend serving **162+ organic users** with **1.8M req/hr** capacity and **99.9% availability** via fault-tolerant fallback on zero paid infrastructure
- **Production Observability:** Implemented Google Analytics 4 tracking **27-min avg session time** and diagnostic logging with **100% test pass rate** under k6 load testing
- **Authentication & UX:** Integrated 1-Click Google OAuth, Resend OTP email verification with 10-min TTL, and password reset flow with **90%+ user retention**
- **Stack:** React (Vite), Node.js, Express, MongoDB, Redis, Cloudflare R2, JWT, Google Analytics 4

### Aura AI — Multilingual AI Image Generator [\[Live\]](#) [\[GitHub\]](#)

- Built distributed AI pipeline supporting **11 Indian languages** with **100% language accessibility** — integrated Gemini API (translation) and Cloudflare AI (image generation)
- Achieved **100 concurrent request handling** with fault-tolerant fallback paths across multi-tiered architecture using ImageKit and MongoDB
- Implemented JWT authentication, rate limiting, and user dashboard — ensuring secure, fault-tolerant multi-user access
- **Stack:** React, Node.js, Express, MongoDB, Gemini API, Cloudflare AI, ImageKit, JWT

### Expense Splitter — Smart Group Settlement System [\[GitHub\]](#)

- Designed debt settlement system using **greedy graph optimization** reducing **50 transactions to 3-5 settlements** ( $O(N^2) \rightarrow O(N-1)$ ) with Strategy Pattern
- **80% manual entry reduction** via Gemini Vision API OCR receipt scanning auto-extracting items, amounts, tax, and GST
- Built interactive dashboard with Recharts and PostgreSQL relational database schema — demonstrating multi-tiered system design and data persistence
- **Stack:** React, Node.js, PostgreSQL, Gemini Vision API, Redis, JWT, Recharts

## LEADERSHIP EXPERIENCE

### Vice Chairperson — IT Students' Association (ITSA), I<sup>2</sup>IT Pune July 2025 – Aug 2026

- Led and mentored a team of 50 members in an Agile environment, fostering a culture of innovation and collaboration while demonstrating Ownership
- Organized multiple events with **200+ student participants** — showcasing collaborative and self-driven leadership

## ACHIEVEMENTS

- **1st Place** — Dhruva Nexus Hackathon (Built internship portal under competition conditions)
- **NoteShare** — 162+ organic users, 27-min avg session time, zero marketing spend
- **Performance** — 77x latency improvement (315ms → 4ms), 1,500+ RPS, 90% DB load reduction

## TECHNICAL SKILLS

|                |                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages      | Java, Python, C/C++, JavaScript, SQL, HTML5, CSS3                                                                                                                                                           |
| Frontend       | React.js, Vite, Tailwind CSS, React Router, Recharts                                                                                                                                                        |
| Backend        | Node.js, Express.js, REST APIs, Microservices                                                                                                                                                               |
| Databases      | MongoDB, PostgreSQL, Redis                                                                                                                                                                                  |
| Cloud & DevOps | AWS, Cloudflare R2, ImageKit, Vercel, Render, Docker, CI/CD                                                                                                                                                 |
| AI/ML          | Gemini API, Cloudflare AI                                                                                                                                                                                   |
| Concepts       | Distributed Systems, Scalability, Fault-Tolerance, High Availability, Multi-level Caching, OOP, Design Patterns, DSA, Complexity Analysis, JWT Auth, RBAC, Agile, Performance Monitoring, Security, Testing |